

SHUBHODEEP GHOSH

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SUMMARY:

Analytical and outcome-oriented Computer Science student with strong foundations in software engineering, data pipelines, and scalable systems. Proficient in Python, SQL, PyTorch, and data visualization, with a proven track record of delivering impactful solutions through machine learning, analytics, and automation. Adept at collaborating across cross-functional teams to solve complex problems in dynamic environments. Passionate about combining data-driven decision making with system design to drive measurable outcomes.

PROJECT EXPERIENCE:

DYNAMIC MICROSERVICE LOAD BALANCER | FastAPI, Redis, Docker

University Project | Jan 2025 – Mar 2025 | Kolkata, India

- Engineered a real-time load balancer to dynamically route service traffic using live CPU/RAM metrics.
- Optimized latency by 40% via intelligent adaptive queuing and Docker-based isolation.
- Deployed Redis-backed routing to handle 10K+ concurrent job requests with seamless failover recovery.

DECENTRALIZED KYC VERIFICATION WITH ZK-STARKS | Python, Solidity, Blockchain, Cryptography

Capstone Project | Feb 2025 – Present | Kolkata, India

- Engineered a privacy-centric KYC framework using zero-knowledge STARKs to verify identities without exposing sensitive data.
- Integrated biometric hashing and smart contracts on blockchain for secure, GDPR-compliant credential validation.
- Simulated multi-node authentication flows reducing onboarding time by ~80% and enabling seamless compliance across institutions.

KRISHISEVAK: AI CROP DIAGNOSIS WEB APP | FastAPI, React, TensorFlow, GPT-4

Team Lead | Dec 2024 – Feb 2025 | Kolkata, India

- Architected a multilingual web platform for real-time plant disease diagnosis via image upload and ML.
- Integrated GPT-4 to generate personalized treatment guidance based on weather and soil API inputs.
- Orchestrated team of 5; achieved 1st rank in university innovation showcase among 50+ teams.

EDUCATION:

Bachelor of Technology – Computer Science and Engineering

University of Engineering and Management, Kolkata

September 2022 - May 2026 | CGPA: 9.04 / 10

Relevant Courses: Data Structures, DBMS, Operating Systems, Machine Learning, Statistics, Algorithms

LEADERSHIP & ACTIVITIES:

Project Team Lead

- Directed multiple end-to-end tech projects, managing stand-ups & sprint reviews to ensure timely delivery.
- Mentored juniors on Python, SQL & project structuring; guided 5+ teams in college hackathons.

Winner, College Software Hackathon

- Secured 1st place for Dynamic Load Balancer project among 50+ teams, judged on innovation & scalability.

ADDITIONAL INFORMATION:

Languages: Python, SQL, Bash

Libraries/Frameworks: pandas, NumPy, scikit-learn, Matplotlib, PyTorch, FastAPI, Hugging Face

Tools: Git, Docker, Google Colab, VS Code

Concepts: Machine Learning, Statistical Analysis, A/B Testing, Microservices, System Design, Data Cleaning

Soft Skills: Problem Solving, Collaboration, Analytical Thinking, Ownership, Strategic Communication